# Execution evidence

Generated by `pnpm report` (`web/scripts/report.ts`) at 2026-10-20T13:56:03.301Z from commit `6df402e429ac0ad8a16b1dcfaadab369639d08c4`. Do not edit by hand.

Every row below is a real transaction on Base Sepolia (chain id 84532), fetched again from the chain while this file was written: status, block, gas and the receiving address come from the receipt, not from the engine. Amounts are demo units with no monetary value.

## 1. Contracts

Deployed from `0x1b9D0C2A0023d21cB9Cf9e20F7bD5d7DFb452d59` in block 48,335,911, source verified on Sourcify with an exact match.

| Contract | Address | Deploy transaction | Gas |
|---|---|---|---|
| IntentLog | 0x093eD0DD8E4bE4b0205a6e39d0c5988f5Cd8ad51 | 0xb43b1cfa…67e9 | 197,751 |
| Bond | 0x6161f6e08c8695849e7A8aF74c39a5A1056d5435 | 0x68f7117a…c578 | 1,949,523 |
| Lease | 0x0D21c8f6C1e6e6EA2ae5fF073F51fC4d57AC5ed2 | 0xeaa13a76…5fdb | 800,050 |
| JobBoard | 0xAbE83C0dD54EAd67DB63C8f9e854f00756f29952 | 0x97227e36…6c1e | 1,517,852 |

## 2. A real run of the fleet scenario

Captured from the public app in real mode (`real-run-2026-10-20.sse`): 339 events, 15 signed intents, 15 policy decisions (2 denied), 7 inbound messages through the gate (0 refused or held), 1 flight log held by the oracle, 1 proof submitted, 40 transactions. Every intent hash goes to the deployed IntentLog before the policy engine decides: 15 were new commits by contract call. Every money event was anchored on the same chain with the event as calldata. No model key was set for this run, so all 68 explanations are labelled template text; section 2b is a run with the free model on.

| # | Seconds | Event | Kind | Sent to | Status | Block | Gas | Events | Transaction |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 5.4 | anchor: vendor bond of 5000 yuan posted | vendor_bond | demo wallet (calldata anchor) | success | 48,336,959 | 28,880 | 0 | 0xd8aa0ef5…5f8f |
| 2 | 15.1 | anchor: fee 60 yuan locked for Spray 20 mu of maize, field north | lock | demo wallet (calldata anchor) | success | 48,336,961 | 29,520 | 0 | 0x15513f3a…c833 |
| 3 | 15.2 | IntentLog.commit: int-001 | log | IntentLog | success | 48,336,962 | 69,288 | 1 | 0x8da01c4b…3ea7 |
| 4 | 15.2 | anchor: job cancelled, fee 60 yuan back to the farmer | refund | demo wallet (calldata anchor) | success | 48,336,963 | 29,160 | 0 | 0x0beb01cf…1bc0 |
| 5 | 21.5 | anchor: fee 1500 yuan locked for Spray 100 mu of passion fruit, field north | lock | demo wallet (calldata anchor) | success | 48,336,963 | 30,040 | 0 | 0x7684af9b…0a86 |
| 6 | 21.6 | IntentLog.commit: int-002 | log | IntentLog | success | 48,336,964 | 69,288 | 1 | 0x556e4442…ceb6 |

| # | Seconds | Event | Kind | Sent to | Status | Block | Gas | Events | Transaction |
|---|---|---|---|---|---|---|---|---|---|
| 7 | 21.6 | IntentLog.commit: int-003 | log | IntentLog | success | 48,336,965 | 69,288 | 1 | 0x67cad212…c459 |
| 8 | 21.6 | IntentLog.commit: int-004 | log | IntentLog | success | 48,336,966 | 69,288 | 1 | 0x700e073d…91d7 |
| 9 | 21.7 | anchor: charging 60 yuan for the flight | energy | demo wallet (calldata anchor) | success | 48,336,966 | 28,600 | 0 | 0x75bcc59b…dd27 |
| 10 | 29.5 | IntentLog.commit: int-005 | log | IntentLog | success | 48,336,966 | 69,288 | 1 | 0xf18f47b9…1f45 |
| 11 | 29.6 | anchor: fee 1500 yuan released to the drone wallet | fee_in | demo wallet (calldata anchor) | success | 48,336,967 | 29,120 | 0 | 0x11adb62e…bfb9 |
| 12 | 29.6 | anchor: rail fee 30 yuan | rail | demo wallet (calldata anchor) | success | 48,336,968 | 27,920 | 0 | 0xe60d0448…374e |
| 13 | 29.7 | anchor: hold 199.95 yuan kept for 24 hours | hold | demo wallet (calldata anchor) | success | 48,336,969 | 28,800 | 0 | 0x368c1d93…f193 |
| 14 | 29.7 | IntentLog.commit: int-006 | log | IntentLog | success | 48,336,969 | 69,288 | 1 | 0x551e1013…2579 |
| 15 | 29.8 | anchor: instalment 600 yuan to the lender | instalment | demo wallet (calldata anchor) | success | 48,336,969 | 28,880 | 0 | 0x7a6065af…396c |
| 16 | 29.8 | anchor: pilot bond filled by 150 yuan from the fee | bond_fill | demo wallet (calldata anchor) | success | 48,336,969 | 29,200 | 0 | 0xcafcd624…e126 |
| 17 | 30.2 | anchor: fee 250 yuan locked for Spray 50 mu of rice, field north | lock | demo wallet (calldata anchor) | success | 48,336,970 | 29,560 | 0 | 0x8068524b…21e7 |
| 18 | 30.3 | IntentLog.commit: int-007 | log | IntentLog | success | 48,336,970 | 69,288 | 1 | 0xa4ae42f0…65e2 |
| 19 | 30.3 | IntentLog.commit: int-008 | log | IntentLog | success | 48,336,971 | 69,288 | 1 | 0x014958f3…e324 |
| 20 | 31.3 | IntentLog.commit: int-009 | log | IntentLog | success | 48,336,972 | 69,288 | 1 | 0xd05a0216…c863 |
| 21 | 32.4 | anchor: charging 30 yuan for the flight | energy | demo wallet (calldata anchor) | success | 48,336,973 | 28,600 | 0 | 0x69f9e7a4…fd4f |
| 22 | 34.7 | anchor: fee 250 yuan held after a flag | refund | demo wallet (calldata anchor) | success | 48,336,974 | 28,600 | 0 | 0xc20a25fe…a5a8 |

| # | Seconds | Event | Kind | Sent to | Status | Block | Gas | Events | Transaction |
|---|---|---|---|---|---|---|---|---|---|
| 23 | 36.7 | IntentLog.commit: int-010 | log | IntentLog | success | 48,336,975 | 69,288 | 1 | 0x2736e598…6a85 |
| 24 | 41.0 | anchor: vendor bond slashed 200 yuan for PayeeNotAllowlisted, intent int-010 | slash | demo wallet (calldata anchor) | success | 48,336,976 | 31,880 | 0 | 0x0f4e1292…8b43 |
| 25 | 41.1 | anchor: 100 yuan to the party that proved it, 100 yuan to the treasury | compensation | demo wallet (calldata anchor) | success | 48,336,977 | 31,920 | 0 | 0x7ec3a417…8853 |
| 26 | 50.1 | anchor: fee 240 yuan locked for Spray 60 mu of rice, field north | lock | demo wallet (calldata anchor) | success | 48,336,977 | 29,560 | 0 | 0xa5fbb062…b3bf |
| 27 | 50.2 | IntentLog.commit: int-011 | log | IntentLog | success | 48,336,978 | 69,288 | 1 | 0x0274b4b6…dd0e |
| 28 | 50.2 | IntentLog.commit: int-012 | log | IntentLog | success | 48,336,978 | 69,276 | 1 | 0x31290d46…224d |
| 29 | 50.3 | IntentLog.commit: int-013 | log | IntentLog | success | 48,336,978 | 69,288 | 1 | 0xad7e9faf…4e57 |
| 30 | 50.3 | anchor: charging 36 yuan for the flight | energy | demo wallet (calldata anchor) | success | 48,336,979 | 28,600 | 0 | 0x28c8b76d…bb43 |
| 31 | 52.8 | IntentLog.commit: int-014 | log | IntentLog | success | 48,336,979 | 69,288 | 1 | 0x9881448b…1621 |
| 32 | 52.8 | anchor: fee 240 yuan released to the drone wallet | fee_in | demo wallet (calldata anchor) | success | 48,336,980 | 29,040 | 0 | 0xfa4f4636…2827 |
| 33 | 52.8 | anchor: rail fee 4.8 yuan | rail | demo wallet (calldata anchor) | success | 48,336,981 | 28,000 | 0 | 0x246306ab…0803 |
| 34 | 52.9 | anchor: hold 31.99 yuan kept for 24 hours | hold | demo wallet (calldata anchor) | success | 48,336,982 | 28,720 | 0 | 0x0bfc4c62…f7d6 |
| 35 | 52.9 | IntentLog.commit: int-015 | log | IntentLog | success | 48,336,982 | 69,288 | 1 | 0x334eb048…9e20 |
| 36 | 52.9 | anchor: instalment 50 yuan to the lender | instalment | demo wallet (calldata anchor) | success | 48,336,983 | 28,800 | 0 | 0xc3a1ba0b…8a24 |
| 37 | 55.0 | anchor: pilot bond filled by 24 yuan from the fee | bond_fill | demo wallet (calldata anchor) | success | 48,336,984 | 29,120 | 0 | 0xb317f69f…804d |

| # | Seconds | Event | Kind | Sent to | Status | Block | Gas | Events | Transaction |
|---|---|---|---|---|---|---|---|---|---|
| 38 | 55.0 | anchor: title transferred from the lender to the pilot | title | demo wallet (calldata anchor) | success | 48,336,983 | 29,120 | 0 | 0xb9d88b29…9880 |
| 39 | 55.2 | anchor: hold 199.95 yuan released, no flag | release | demo wallet (calldata anchor) | success | 48,336,984 | 28,920 | 0 | 0x4a90a4c2…943e |
| 40 | 57.3 | anchor: hold 31.99 yuan released, no flag | release | demo wallet (calldata anchor) | success | 48,336,985 | 28,840 | 0 | 0x77971312…d461 |

Checked 40 hashes against the chain: 40 succeeded, 0 not found, 15 were `IntentLog.commit` calls with a `Committed` event each, 1,768,708 gas in total (0.0000106 ETH at 0.006 gwei).

## 2b. A second real run: free model and FISCO mirror on

`real-run-2026-10-20T21.sse`, captured from the public app with the demo token after the free model and the FISCO target were switched on: 336 events, 39 transactions, 39 of 39 receipts fetched again now with status success. 5 explanations were written by a free model on OpenRouter and passed the number check, 65 are labelled template text (about ten model calls a minute on the free tier). 15 intents carry a FISCO BCOS hash next to the Base one. The intent hashes were already in the Base IntentLog from the run above, so on Base they were anchored as calldata; `pnpm verify:all` re-checks every one of these receipts.

## 3. One full job against the deployed contracts

Run by `contracts/script/fulljob.sh base`: fresh keys for six roles, every step a call to the deployed Bond, Lease, JobBoard or IntentLog, value moving where the contract moves it. These are contract calls, not calldata anchors. 9 of 9 receipts fetched again now with status success. Amounts are a few gwei, enough to prove the flow at almost no cost.

| # | Call | From | Value | Status | Block | Gas | Transaction |
|---|---|---|---|---|---|---|---|
| 1 | Bond.postVendorBond | vendor | 20000000000 wei | success | 48,365,739 | 90,669 | 0xf080cce8…d8f6 |
| 2 | Bond.register | agent | 0 | success | 48,365,741 | 162,649 | 0x5218a105…c53d |
| 3 | Lease.open | lender | 0 | success | 48,365,744 | 192,409 | 0x8a000651…f04d |
| 4 | JobBoard.post | farmer | 100000000000 wei | success | 48,365,747 | 187,428 | 0xc1d373ee…d899 |
| 5 | JobBoard.start | agent | 0 | success | 48,365,749 | 46,171 | 0x3f708c8b…954f |
| 6 | JobBoard.confirm | verifier | 0 | success | 48,365,753 | 176,680 | 0x1b3f71b8…7db6 |
| 7 | IntentLog.commit | agent | 0 | success | 48,365,760 | 68,916 | 0x05af20e0…c73c |
| 8 | Bond.slashByProof | lender | 0 | success | 48,365,763 | 187,021 | 0x68606a4c…d748 |
| 9 | Lease.pay | pilot | 60000000000 wei | success | 48,365,767 | 66,955 | 0x5440d994…3bd3 |

What moved, read from balances at each transaction's block:

- vendor bond after post (wei): 20000000000 (before 0)
- operatorOf(agent) is pilot: 0x617cB14a84A40021E26A907b9edF9e9791d2060d
- lease id: 1

- title owner after open (lender): 0x55AFCCF107e322e0A040Ead76b475b7788D4C4Ab
- job id: 1
- JobBoard balance delta after post (fee locked): 100000000000
- lender received instalment via Lease.pay (wei): 40000000000 (balance at the tx block minus the block before; the first read through the public RPC was stale and showed 0)
- agent received rest (wei): 34670000000
- pilot bond filled (wei): 10000000000
- hold kept in JobBoard (wei): 13330000000
- lease balance after confirm (wei): 60000000000
- rogue intent hash: 0x45229be9c73782e18be195bc2cbea7557063093d338b40dc80ba47b5c45d4017
- rogue intent signer recovered off chain: agent
- vendor bond decreased by (wei): 10000000000
- treasury (deployer) received penalty (wei): 10000000000
- lender received remaining lease (wei): 60000000000 (balance at the tx block minus the block before; the first read through the public RPC was stale and showed 0)
- lease balance after final payment (wei): 0
- title owner after zero balance: 0x617cB14a84A40021E26A907b9edF9e9791d2060d
- hold: releaseHold callable after holdUntil=1792586194 (one day after confirm), not called in this run
- rail (deployer) received rail fee (wei): 2000000000 (balance at the confirm block minus the block before)

The same job ran on FISCO BCOS, see `contracts/deployments/fisco-bcos.json` and `docs/FISCO_EVIDENCE.md`. The funding and sweep transfers for the role wallets are listed in `contracts/deployments/base-sepolia-fulljob.json`.

## 4. Reproduce

```
cd web && pnpm install
pnpm test                       # 52 attacks blocked, 7 benign pass, every learned metric at or abov
e its floor
pnpm evidence && pnpm verify    # a simulated run and its balance sheet, every hash checked
# real chain: a funded Base Sepolia key and the demo token in the environment, then
JIXIN_REAL=1 pnpm evidence      # anchors on Base Sepolia
pnpm report                     # rewrites this file from the latest capture and the chain
```

Any hash above can be checked on Basescan or with one RPC call:

```
curl -s https://sepolia.base.org -H 'content-type: application/json' -d '{"jsonrpc":"2.0","id":1,"me
thod":"eth_getTransactionReceipt","params":["<hash>"]}'
```

## 5. What is simulated

In the live engine the farmer, the pilot, the lender and the vendor are scripted actors with software keys (random per server instance in the deployed app; the captured runs used the fixed test keys so their intent hashes reproduce). Flight telemetry is synthetic. On Base Sepolia the live engine calls IntentLog and anchors money events (the funded test wallet is small); section 3 is the full flow against the deployed contracts here, and on FISCO BCOS the engine itself settles every money event on the deployed contracts (docs/FISCO_EVIDENCE.md

section 8). The public app runs on FISCO when its node is reachable and on the labelled simulation otherwise; Base Sepolia real mode needs the demo token. The captured run had no model key set, so every explanation in it is the labelled template text (source template); decisions never depend on the model. With a key the model rewords explanations and the ask the agent replies, checked against the facts.